



What is React?



JavaScript Library

A lightweight, focused tool for building UIs



Facebook Developed

Created and maintained by Meta (Facebook)



Dynamic Updates

Updates content without reloading the entire page

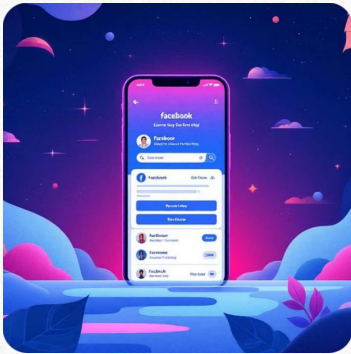


Component-Based

Builds interfaces using reusable components

Real-Life Examples

Major platforms like Facebook, Instagram, and YouTube leverage React to deliver seamless experiences. These applications update content dynamically without full page reloads, providing users with a fluid, app-like experience.



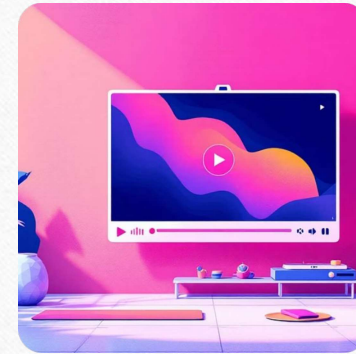
Facebook

News feed, comments, and reactions update instantly



Instagram

Photo feeds and stories load seamlessly



YouTube

Video recommendations and comments refresh without reloading

Virtual DOM: The Performance Advantage

How It Works

React creates a **Virtual DOM**, which is an in-memory representation of the real DOM.

When data/state changes, React first updates the **Virtual DOM** instead of the real DOM.

React then compares the updated Virtual DOM with the previous version using a process called **diffing**.

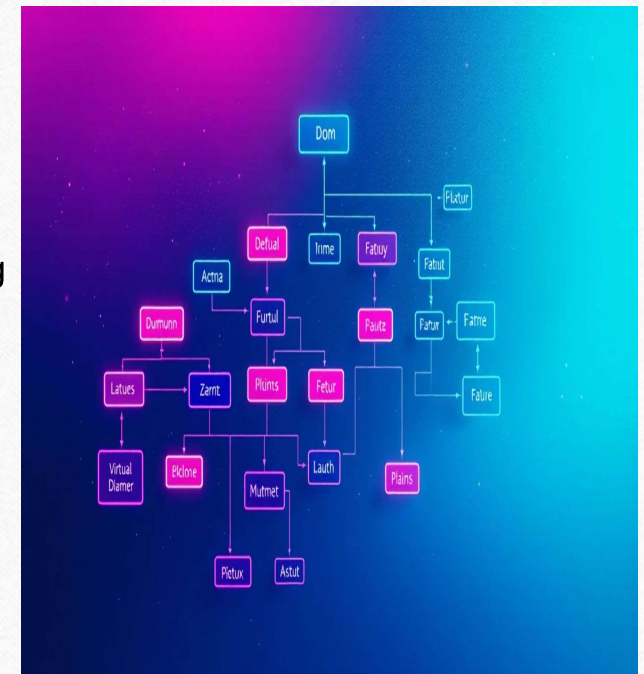
It identifies only the elements that have changed.

React updates only those specific parts in the **real DOM**.

This minimizes unnecessary DOM manipulations.

As a result, the application becomes **faster and more efficient**.

This approach improves overall **performance and rendering speed**



Component-Based Architecture

React breaks down complex user interfaces into smaller, manageable, and reusable components. Each component is self-contained, managing its own state and logic whilst maintaining a clear structure.



Navbar

Navigation menus that stay consistent across all pages



Footer

Site footer with contact information and links



Button

Interactive buttons with consistent styling and behaviour



Card

Content containers for articles, products, or user profiles



Single Page Application (SPA)

01

One Page Load

The entire application loads once without refreshing

02

Dynamic Content

Only required content updates when navigating

03

Smooth Experience

Provides fluid, app-like user experience

Why Choose React?

High Performance

Virtual DOM ensures lightning-fast updates and rendering

Easy to Learn

Beginner-friendly syntax with JavaScript and JSX

Reusable Components

Write once, use anywhere across your application

Scalable Architecture

Perfect for managing large, complex applications

Strong Community

Vast ecosystem of tools, libraries, and resources

High Demand

Extremely sought-after skill in the job market

Traditional JavaScript vs React

Traditional JavaScript

- Full page reload on every update
- Slower performance
- Less structured code
- Manual DOM manipulation

React

- No full page reload
- Faster performance
- Well-structured component code
- Automatic efficient updates



Final Summary

JavaScript Library

A focused tool for building modern UIs

Component Architecture

Reusable, self-contained building blocks

Virtual DOM

Efficient updates without full reloads

Excellent UX

Fast, smooth, app-like experience

Installation Steps (React Setup using Vite)

Follow these straightforward steps to set up your first React project using Vite, a blazing-fast build tool that significantly improves the development experience.

01

Install Node.js

Download and install Node.js from the official website, which includes npm (Node Package Manager).

02

Open Your Terminal

Launch your preferred terminal or command prompt application.

03

Create React App

```
npm create vite@latest.
```

```
cd project-name
```

```
npm install
```

```
npm run dev
```

04

View Your Application

Open the provided local URL

(e.g., `http://localhost:5173/`)

in your web browser to see

your React application.

Note :- it was old way to create project
`npx create-react-app my-app`